An internal block action in the remaining support branch

Human Compression project: bounded L044 proof component

Let $G = \langle \rho_0, \dots, \rho_{r-1} \rangle \le A_n$ be transitive, $n \ge 12$, with a connected string C-marking and every maximal omission intransitive. Let the unique eligible i-edge $\{a, b\}$, $0 < i < r-1$, join the two G_i-orbits A, B. Assume $|B| \ge 2$ and $L = G_{<i}$ fixes B pointwise and is intransitive on A. Write $\beta = a^L$, $k = |\beta|$, and $\tau = \rho_{i+1}$.

The L-orbits in A are blocks for $G_i = LG_{>i}$, because the two factors commute. They have equal sizes. Every ρ_j with $j \ge i+2$ fixes a and centralizes L, so fixes β pointwise. Thus τ must send β to a different block β', otherwise β would be G_i-invariant. Every generator of index at least $i+3$ also fixes β' pointwise, by commutation with τ. The involution τ moves b as well: it is the only member of G_i that can do so, and B is a nontrivial G_i-orbit.

Lemma 1 (A faithful commuting-orbit action)**.** *The action of L on each of its orbits in A is faithful, and the actions on β and β' are identified by τ. Suppose $\sigma = \rho_{i+2}$ exists, preserves β', and acts nontrivially there. Then $A = \beta \cup \beta'$, k is even, and*

$$i \le k/2 - 1. \tag{1}$$

Proof. A member of L fixing one L-orbit pointwise fixes all its translates under $G_{>i}$, by commutation. Those translates cover A; it also fixes B, hence is the identity. This proves faithfulness and the asserted identification.

The set $\beta \cup \beta'$ is preserved by L, interchanged blockwise by τ, preserved by σ, and fixed pointwise by every later generator. It is therefore all of A. The restriction of σ to β' centralizes the transitive L-action. Being nontrivial and involutory, it has no fixed point, so supplies $k/2$ pairs. The conjugate σ^τ supplies corresponding pairs on β. These are L-invariant block systems.

Choose a minimal distinguished subset M of L generating its whole action on these $k/2$ pairs. This block action is transitive, so $\langle M \rangle$ has at most two orbits in β. If there are two, σ^τ interchanges them; hence $\langle M, \sigma^\tau \rangle$ is transitive on β in either case. Similarly $\langle M, \sigma \rangle$ is transitive on β'. Together with τ these groups are transitive on A.

If a distinguished L generator ρ_j were outside M, all the displayed elements would belong to G_j, so G_j would be transitive on A. But ρ_j is supported on A, and could not join distinct G_j-orbits. This contradicts transitivity of $G = \langle G_j, \rho_j \rangle$ and intransitivity of G_j. Thus every L generator belongs to M. Their pair images are an independent family in $S_{k/2}$; the retained Whiston bound gives (1). Since $i > 0$, necessarily $k \ge 4$. No projected intersection property is used. □

Proposition 2 (Strict support deficit or completed rank bound)**.** *Under the internal-block hypothesis of the lemma, if $i+2 = r-1$, then $r \le \lfloor (n-1)/2 \rfloor$. Otherwise $h = i+2$ is the largest shared label, and with $X = \mathrm{supp}(G_{>h})$ one has*

$$X \subseteq B \setminus \{b\}, \qquad |\operatorname{Fix}(G_{>h})| \ge 2k+1, \qquad h \le \frac{n - |X| - 3}{2}.$$

In particular the equality support configuration in source Proposition 5.23 cannot arise in this subcase.

Proof. If $i+2 = r-1$, (1) and $n \geq 2k+2$ give

$$r = i+3 \leq k/2+2 \leq k \leq \left\lfloor \frac{n-1}{2} \right\rfloor,$$

using $k \geq 4$.

If a later generator exists, ρ_{i+3} fixes A and acts nontrivially on B. Were σ trivial on B, their supports would be disjoint, so these adjacent generators would commute. All generators on opposite sides of this adjacent pair already commute by the string condition; this would disconnect the noncommutation diagram. Hence σ also acts nontrivially on B. It is the last shared generator because all later generators fix A. The tail also fixes b, by the unique-crossing lemma. It fixes at least $2k+1$ points, while (1) gives

$$2h+3 = 2i+7 \leq k+5 \leq 2k+1$$

for $k \geq 4$. This proves the strict bound and the support placement claimed. □

Scope and attribution. This is the internal-block subcase of Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Proposition 5.23, physical page 32. The faithful commuting-orbit and actual-omission argument makes its rank reduction explicit, using the paid unique-crossing proof and retained Whiston bound. The application to $G = A_n$ has a connected diagram by simplicity; for a general subgroup the connectedness assumption is retained. The nontrivial-B hypothesis is explicit in this successor: being the only possible mover of b alone does not exclude a singleton orbit. A shared generator in the intended source branch ensures $|B| \geq 2$. No finite computation or formal theorem is added. The case where ρ_{i+2} moves β' to another block, all further path/equality rigidity and the whole L044 contract remain separate.